

Definition 0.1 (Boolean function): Let $\mathbb{F}_2$ denote the finite field of two elements (isomorphic to $\mathbb{Z}/2\mathbb{Z}$). A Boolean function is a map:

$$f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$$

The alternate form maps the multiplicative group of units $\{-1, 1\}$ (isomorphic to the additive group $\mathbb{F}_2$ via $x \mapsto (-1)^x$) to itself:

$$f : \{-1, 1\}^n \rightarrow \{-1, 1\} \subset \mathbb{R}$$

Definition 0.2 (Hamming cube): The n -dimensional Hamming cube, denoted as Q_n or H_n , is the vector space $\mathbb{F}_2^n$. It consists of all n -tuples of elements from $\mathbb{F}_2$:

$$\mathbb{F}_2^n = \{(x_1, \dots, x_n) \mid x_i \in \mathbb{F}_2\}$$

Definition 0.3 (Hamming distance): Let $x, y \in \mathbb{F}_2^n$. The Hamming distance is the [metric] $d_H : \mathbb{F}_2^n \times \mathbb{F}_2^n \rightarrow \mathbb{N} \cup \{0\}$ defined by the Hamming weight of the difference vector:

$$d_H(x, y) = \|x - y\| = wt(x + y)$$

where addition is performed in $\mathbb{F}_2$ (bitwise XOR) and $\|v\|$ (or $wt(v)$) denotes the number of non-zero components in the vector v .

0.0.1 Fourier expansion of Boolean functions

Note 1: From here on, it is convenient to switch the domain from $\mathbb{F}_2^n$ to $\{-1, 1\}^n$. We consider real-valued functions $f : \{-1, 1\}^n \rightarrow \mathbb{R}$.

To analyze Boolean functions using Fourier analysis, we view the set of all functions $f : \{-1, 1\}^n \rightarrow \mathbb{R}$ as a vector space of dimension 2^n , denoted as $L^2(\{-1, 1\}^n)$. We equip this space with an inner product defined by the expectation over the uniform distribution.

Definition 0.4 (Inner Product): For functions $f, g : \{-1, 1\}^n \rightarrow \mathbb{R}$, the inner product is defined as:

$$\langle f, g \rangle := \mathbb{E}_{x \sim \{-1, 1\}^n} [f(x)g(x)] = \frac{1}{2^n} \sum_{x \in \{-1, 1\}^n} f(x)g(x)$$

We also use the notation $\|f\|_2 := \sqrt{\langle f, f \rangle}$, and more generally

$$\|f\|_p := \mathbb{E} [|f(x)|^p]^{1/p}.$$

The Fourier basis for this space consists of the parity functions (or characters). While often indexed by subsets $S \subseteq [n]$, we can equivalently index them using vectors $S \in \mathbb{F}_2^n$.

Definition 0.5 (Character Functions): For a subset indicator $S \in \mathbb{F}_2^n$, the character function $\chi_S : \{-1, 1\}^n \rightarrow \{-1, 1\}$ is defined as:

$$\chi_S(x) := \prod_{i: S_i=1} x_i$$

Note that $\chi_{\mathbf{0}}(x) \equiv 1$ (where $\mathbf{0}$ is the zero vector).

Lemma 0.6 (Orthonormality): *The set of functions $\{\chi_S\}_{S \in \mathbb{F}_2^n}$ forms an orthonormal basis for the space of real-valued functions on the hypercube. That is:*

$$\langle \chi_S, \chi_T \rangle = \begin{cases} 1 & \text{if } S = T \\ 0 & \text{if } S \neq T \end{cases}$$

Theorem 0.7 (Fourier expansion of Boolean functions): *Every function $f : \{-1, 1\}^n \rightarrow \mathbb{R}$ can be uniquely expressed as a linear combination of the character functions:*

$$f(x) = \sum_{S \in \mathbb{F}_2^n} \hat{f}(S) \chi_S(x)$$

The real numbers $\hat{f}(S)$ are called the Fourier coefficients of f and are given by projecting f onto the basis functions:

$$\hat{f}(S) = \langle f, \chi_S \rangle = \mathbb{E}_{x \sim \{-1, 1\}^n} [f(x) \chi_S(x)]$$

Theorem 0.8 (Parseval's Identity): *For any $f : \{-1, 1\}^n \rightarrow \mathbb{R}$, the energy of the function is preserved in the Fourier domain:*

$$\langle f, f \rangle = \mathbb{E}_x [f(x)^2] = \sum_{S \in \mathbb{F}_2^n} \hat{f}(S)^2$$

For Boolean functions where $f(x) \in \{-1, 1\}$, it follows that $\sum \hat{f}(S)^2 = 1$.